



Project: Group Formation

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Due: Friday, January 23rd

In place of a final exam, this course has a collaborative final project where we ask you to bring your data science skills to bear on a research project of your own choosing. You'll submit your group preferences now (teams of 4-5 students). **Fill out the group formation quiz on Gradescope no later than Friday, January 23rd.**

The form should only take a couple of minutes. If you already know who you want to work with, you can indicate that in the form using the [group submission feature in Gradescope](#). In this case, **communicate with your group first and have one member fill out the form once with everyone added as group members**. If you submit more than once, the active submission is considered valid. It's also fine if you don't know who you want to work with, in which case you should fill out the form solo, and we will match you to a group.

If it is helpful to start thinking about possible project ideas, below are some examples of what a project might look like. You can also brainstorm now using

strategies that are outlined in the Initial Plan post (out soon). But it is not required that you have a concrete project idea until the proposal.

Project Ideas

Not sure how to get started? Looking for examples of what a data science project might look like? Here are some of the topics that students studied in Spring 2020:

- Comparing Stock Market Losses between SARS and SARS-CoV-2
- Recessions, Depressions, and Depression: Mental Health in Relation to Economic Factors
- Predicting North Carolina Election Outcomes
- Relating Text Analysis of Corporate Reports and Stock Performance
- Modeling Consumer Flight Behavior Based on Economic Indicators
- Predicting COVID-19 Death Tolls from Google Search Trends
- Sentiment Analysis of COVID-19 Tweets
- Economic Status and Drug Overdose in North Carolina
- Analyzing Gender and Tech Careers
- Political Landscape According to Social Media
- Forecasting Market Shocks and Performance using Article Headlines
- Tracking Recidivism in US Prisons
- Understanding AirBnBs impact on Evictions
- Understanding Musical Tastes (Music Recommender System)
- Human Impact on Climate since the Industrial Revolution
- The Troll Toll: An Investigation into Troll Tweets

And here is an [archive of summer Data+ projects](#) from the last several years. In Data+, teams of about 4 undergraduate students collaborate over the summer on a data science project. You should be able to see final presentations and/or executive summary slides for most projects; feel free to browse for inspiration.

Example Data Sources

Below, we have some examples of datasets or where you might find data. You should work with data that is interesting to you and should feel free (strongly encouraged even) to look for sources yourself. These are listed just as possibilities and starting places.

- [Data.gov](#) has a huge compilation of data sets produced by the US government. The [US Census Bureau](#) also publishes datasets from all of its survey work. Similarly, The [Supreme Court Database](#) tracks all cases decided by the US Supreme Court, and [GovTrack.us](#) provides links to all kinds of information about the US Congress and all votes casted by its members.
- [Duke University Library Digital Repository Research Data](#) provides access to research datasets produced by the Duke community.
- [ICPSR](#) – An international consortium of more than 750 academic institutions and research organizations, Inter-university Consortium for Political and Social Research (ICPSR) provides leadership and training in data access, curation, and methods of analysis for the social science research community. ICPSR maintains a **data archive** of more than 250,000 files of research in the social and behavioral sciences. It hosts 21 specialized collections of data in education, aging, criminal justice, substance abuse, terrorism, and other fields.
- The University of California Irvine maintains a large [UCI ML repository](#) of publicly contributed datasets aimed toward machine learning tasks of all types. They range from small simple example datasets to large and complicated datasets from specific scientific domains.
- [Kaggle](#) maintains several thousand public datasets of interest in a variety of topics. Kaggle also hosts several prediction challenges; one idea for a machine learning project is to enter one of these competitions as a team. However, watch out for synthetic datasets on this platform that are artificially generated instead of sourced from real data.
- The [Yelp Dataset](#) is provided by Yelp as a research challenge with lots and lots of data about reviews, businesses, images, and cities – text data, rich json data, etc.

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